

Automorphism Representability of $\mathbb{R}$

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A group H is *[group] automorphism-representable* if there exists a group G such that

$$\text{Aut}_{\mathbf{Grp}}(G) \cong H$$

From now on we'll just write Aut instead of $\text{Aut}_{\mathbf{Grp}}$.

We shall prove the following:

Theorem 0.0.1: Torsion-Free Divisible Abelian Groups not Automorphism Representable

Let H be a torsion-free divisible abelian group. Then there does not exist a group G such that

$$\text{Aut}(G) \cong H$$

This result is a generalization of the following simpler result:

Lemma 0.0.2: Rational Are Not Automorphism-representable

The rational numbers under addition $(\mathbb{Q}, +)$ are not automorphism representable

Proof:

The key to remember is that finitely generated subgroups of $\mathbb{Q}$ are cyclic. For the sake of contradiction say they are so that there exists a G such that:

$$\text{Aut}(G) \cong \mathbb{Q}$$

Then as:

$$\text{Inn}(G) \trianglelefteq \text{Aut}(G) \cong \mathbb{Q}$$

either $\text{Inn}(G)$ is a divisible group or cyclic. If it is cyclic then it must have infinite order as $\mathbb{Q}$ is torsion free. Then as:

$$G/Z(G) \cong \mathbb{Z}$$

Elementary group theory tells us G is abelian. But if G is abelian there must be a torsion element $x \mapsto x^{-1}$ of order 2, a contradiction. On the other hand, if $G/Z(G)$ is not finitely generated, we can “cheat” by choosing any $gZ(G), hZ(G) \in G/Z(G)$ and generating the group H which is cyclic, so

$$\langle xZ(G) \rangle = \langle gZ(G), hZ(G) \rangle$$

Then it is easy to see that:

$$gh = x^n z_1 x^m z_2 = z^{n+m} z_1 z_2 = z^m z_2 x^n z_1 = hg$$

and as g, h were arbitrary, G is abelian giving us the same contradiction.

Another important fact: let us make sure that if G is an $\mathbb{F}_2$ -vector-space then $\text{Aut}(G)$ cannot be abelian. The point of this is that we want to abuse the $x \mapsto x^{-1}$ argument, but we can't do that if $x^{-1} = x$ and hence this is the identity:

Lemma 0.0.3: Vector Space, then Non-abelian

Assume $x \mapsto x^{-1}$ is the identity on the group G . Then G has an $\mathbb{F}_2$ -action, hence is a vector space and hence $\text{Aut}(G)$ cannot be abelian

Proof:

Say that $x \mapsto x^{-1}$ is the identity. Then every element has a natural $\mathbb{F}_2$ -action on it. Hence G is a vector-space which we know by linear algebra $\text{Aut}(G) = \text{GL}(G)$ is not abelian.

We shall be clever with such arguments in a moment.

Central automorphism and cohomology

We shall explicitly aim to prove that if A is a torsion-free divisible abelian group (like $\mathbb{Q}$, $\mathbb{Q}^n$, or $\mathbb{R}$), then it is not automorphism representable. For the sake of contradiction let's say $\text{Aut}(G) \cong A$. As A is abelian, we can conclude all automorphisms are *central*:

Definition 0.0.4: Central Automorphisms

An automorphism $\alpha \in \text{Aut}(G)$ is called *central* if it induces the identity map on the central quotient $G/Z(G)$. Equivalently, for all $x \in G$, $x^{-1}\alpha(x) \in Z(G)$. In other words:

$$\alpha(x) \in xZ(G)$$

so α keeps elements in the cosets.

We claimed that:

$$\text{Aut}_c(G) \cong \text{Aut}(G)$$

To show this, since $\text{Aut}(G)$ is abelian, the subgroup of inner automorphisms, $\text{Inn}(G) \cong G/Z(G)$, must be a normal subgroup of $\text{Aut}(G)$. Let $\alpha \in \text{Aut}(G)$ and $c_g \in \text{Inn}(G)$ (where $c_g(x) = gxg^{-1}$). Since $\text{Aut}(G)$ is abelian, α and c_g commute:

$$\alpha \circ c_g = c_g \circ \alpha$$

Applying this to all $x \in G$:

$$\begin{aligned}\alpha(gxg^{-1}) &= g\alpha(x)g^{-1} \\ \alpha(g)\alpha(x)\alpha(g)^{-1} &= g\alpha(x)g^{-1}\end{aligned}$$

Since α is surjective, as x varies over G , $\alpha(x)$ varies over all of G . Letting $y = \alpha(x)$, re-arranging we see that:

$$g^{-1}\alpha(g)y = yg^{-1}\alpha(g)$$

Letting $z = g^{-1}\alpha(g)$ we get:

$$zy = yz$$

Hence, as y varies through all elements of G ,

$$z = g^{-1}\alpha(g) \in Z(G) \quad \text{i.e.} \quad \alpha(g) \in gZ(G)$$

which is exactly the condition of a central automorphism. Thus if $\text{Aut}(G)$ is abelian, then

$$\text{Aut}(G) = \text{Aut}_c(G)$$

We shall abuse this with some cohomology. First, from group theory we know that $\text{Inn}(G) \cong G/Z(G)$ so that:

$$0 \rightarrow Z(G) \rightarrow G \rightarrow \text{Inn}(G) \rightarrow 0$$

is a central extension, where $\text{Inn}(G) \leq A$ is abelian. hence, G is a nilpotent group of order 2, namely $[G, G] \leq Z(G)$. This will allow for great cohomological restraints. Recall that $H^2(Z(G), \text{Inn}(G))$ is in bijective correspondences with all extensions, characterized by the choice of 2-cocycles. Namely, if $Q = G/Z(G)$ and $Z = Z(G)$, then since G is a central extension of Z by Q , elements of G can be identified with pairs (a, z) where $a \in Q$ and $z \in Z$, with the multiplication rule:

$$(a, z_1) \cdot (b, z_2) = (a + b, z_1 + z_2 + f(a, b))$$

where $f : Q \times Q \rightarrow Z$ is a 2-cocycle (factor set); see (cite Neukirch and Robinson and myself). Note that $f(a, b) \neq f(b, a)$ (or else this group would be abelian and we know this implies $\text{Aut}(G)$ has to have an order 2 element), and how we are using the fact that the 2nd term is the center, or else we would have to write:

$$(a, z_1) \cdot (b, z_2) = (a + b, a \cdot z_1 + z_2 + f(a, b))$$

for some action of a on z_1 , but this action is trivial using the center (ibid on sources).

Now comes the crucial part: as G is nilpotent of class 2, we have the following powerful theorem for nilpotent groups: By Warfield (*Nilpotent Groups*, Theorem 5.4), the extension class of G in $H^2(Q, Z)$ maps to a commutator form $\omega \in \text{Hom}(\Lambda^2 Q, Z)$, namely:

$$f(x, y) \mapsto f(x, y) - f(y, x) = \omega(u, v) \in \text{Hom}(\Lambda^2 Q, Z)$$

In his words:

Theorem 0.0.5: Warfield, Thm 5.4

If A and B are Abelian groups, and $H^2(B, A)$ is the group of central extensions of A by B , then there is an exact sequence

$$0 \rightarrow \text{Ext}(B, A) \rightarrow H^2(B, A) \rightarrow \text{Hom}(\Lambda^2 B, A) \rightarrow 0$$

where the map on the right is given by the commutator form. If A is uniquely 2-divisible, the sequence has a canonical splitting. Otherwise, the sequence splits, but not necessarily in a unique way.

The theorem states that ω is an *Alternating Bilinear Form*. As it is bilinear, it implies:

$$\omega(-u, -v) = (-1)(-1)\omega(u, v) = \omega(u, v)$$

0.0.1 Using this result to eliminate some immediate cases

The main things we need is that $A = G/Z(G)$ is at least 2-divisible, since in that case the sequence splits and we can write:

$$f(x, y) = \frac{1}{2} (f(x, y) - f(y, x)) + \frac{1}{2} (f(x, y) + f(y, x))$$

If this is the case, then we can define a homomorphism (clearly bijective, hence isomorphism):

$$\Psi : G \rightarrow G \quad \Psi(a, b) = (-a, b)$$

indeed:

$$\begin{aligned} \Psi((a, z_1) \cdot (b, z_2)) &= \Psi(a + b, z_1 + z_2 + f(a, b)) \\ &= (-(a + b), z_1 + z_2 + f(a, b)) \end{aligned}$$

$$\begin{aligned} \Psi(a, z_1) \cdot \Psi(b, z_2) &= (-a, z_1) \cdot (-b, z_2) \\ &= (-a - b, z_1 + z_2 + f(-a, -b)) \end{aligned}$$

Since $f(a, b) = f(-a, -b)$, the map preserves the group operation. Since $\Psi^2 = \text{id}$, it is bijective and thus an automorphism. Note the important of bi-linearity to insure it is well-defined. It is now an order 2 automorphism, a contradiction .

Similarly, if B or A is free or even projective (so $Z(G)$ or $G/Z(G)$ are free), then $\text{Ext}(B, A) = \{0\}$ and we get that $f(x, y)$ must be an alternating form, and so we get the same argument.

0.0.2 Lie-Algebra case

Next, it may be that A is not 2-divisible, but it *is* still t -divisible for some $t \in \mathbb{N}$, we employ the following equivalence of categories with rational nilpotent lie algebras.

[here](#).

0.0.3 Where the ideas stop working

In the mixed case, say $A = \mathbb{Q} \times \mathbb{Z}$, then I am stuck.

0.0.4 Further ideas

With this, we define a specific map $\Psi : G \rightarrow G$ and show it is an automorphism that violates the constraint established. Let $Q = G/Z(G)$ and $Z = Z(G)$. Since G is a central extension of Z by Q , elements of G can be identified with pairs (a, z) where $a \in Q$ and $z \in Z$, with the multiplication rule:

$$(a, z_1) \cdot (b, z_2) = (a + b, z_1 + z_2 + f(a, b))$$

According to the theory of nilpotent groups (see Warfield, *Nilpotent Groups*, Theorem 5.4), for torsion-free groups, the factor set f corresponds to an alternating bilinear form (the commutator map). Bilinearity implies:

$$f(-a, -b) = (-1)(-1)f(a, b) = f(a, b)$$

We define the map $\Psi : G \rightarrow G$ by:

$$\Psi((a, z)) = (-a, z)$$

But now, we have established two contradictory facts about Ψ :

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1. Since $\Psi \in \text{Aut}(G)$ and $\text{Aut}(G)$ is abelian, Ψ must be a central automorphism. It must induce the identity map on Q .

$$\Psi(a, z) \equiv (a, z) \pmod{Z} \implies -a = a$$

2. By definition, Ψ induces the inversion map on Q .

$$\Psi(a, z) = (-a, z)$$

These conditions require $-a = a$, or $2a = 0$, for all $a \in Q$. Since Q is a subgroup of $\mathbb{Q}^n$, it is *torsion-free*. The only solution to $2a = 0$ is $a = 0$. This implies Q is the trivial group, so $G = Z(G)$, and G is abelian. The abelian case was already established, hence no such group can exist.